

PRACTICAL WEEK 2

Task 1: Develop a C program that uses the system calls `open()`, `read()`, `write()`, and `close()` to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution.

1) Create a file

```
saichaitanya@GeethzzSai:~$ cd pweek2
saichaitanya@GeethzzSai:~/pweek2$ nano task1.c
saichaitanya@GeethzzSai:~/pweek2$ |
```

Then press enter to open command

2) write a code

```
GNU nano 7.2 main.c
#include <fcntl.h>
#include <unistd.h>
#include <stdio.h>
#include <stdlib.h>

int main() {
    int source, destination;
    char buffer[1024];
    ssize_t bytesRead, bytesWritten;

    source = open("source.txt", O_RDONLY);

    if (source < 0) {
        perror("Error opening source file");
        return 1;
    }

    destination = open("destination.txt",
        O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (destination < 0) {
        perror("Error opening destination file");
        close(source);
        return 1;
    }

    printf("Source file opened successfully.\n");
    printf("Destination file opened successfully.\n");

    while ((bytesRead = read(source, buffer, sizeof(buffer))) > 0) {
        bytesWritten = write(destination, buffer, bytesRead);

        if (bytesWritten != bytesRead) {
            perror("Error writing to destination file");
            close(source);
            close(destination);
            return 1;
        }
    }
}
```

```

        close(destination);
        return 1;
    }
}

if (bytesRead < 0) {
    perror("Error reading source file");
} else {
    printf("File contents copied successfully.\n");
}

close(source);
close(destination);

printf("Files closed successfully.\n");

return 0;
}

```

- 1) To save press control O
- 2) To come back to shell press control X
- 3) Then DO compilation

```

saichaitanya@GeethzzSai:~$ cd pweek2
saichaitanya@GeethzzSai:~/pweek2$ nano task1.c
saichaitanya@GeethzzSai:~/pweek2$ gcc task1.c -o task1
saichaitanya@GeethzzSai:~/pweek2$ |

```

OUTPUT:

```

saichaitanya@GeethzzSai: ~/  ×  +  ▾
saichaitanya@GeethzzSai:~$ cd pweek2
saichaitanya@GeethzzSai:~/pweek2$ nano task1.c
saichaitanya@GeethzzSai:~/pweek2$ gcc task1.c -o task1
saichaitanya@GeethzzSai:~/pweek2$ echo "Hello from Task 2 sample file" > sample.txt
saichaitanya@GeethzzSai:~/pweek2$ cat source.txt
Hello, this is a system call demonstration.
This file will be copied using open, read, write and close.

saichaitanya@GeethzzSai:~/pweek2$ ./task1
File copied successfully.
saichaitanya@GeethzzSai:~/pweek2$ |

```

